

LEVEL - VI

SINGLE ANSWER QUESTIONS

1. The sequence of matrices is defined as given below $m(1)=[1]$

$$m(2)=\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, m(3)=\begin{bmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$$

and so on then the trace of $m(17)$ is

- A) 27,895 B) 23,679
C) 27,761 D) 27,987

2. Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ then trace of A^3 is

- A) 101 B) 100 C) 99 D) 98

3. If A and B are symmetric matrices of the same order $X = AB + BA$ and $Y = AB - BA$ then $(XY)^T$ is equal to

- A) XY B) YX
C) $-YX$ D) None of these

4. Let $\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \begin{pmatrix} \sqrt{3}+1 & 1-\sqrt{3} \\ \sqrt{3}-1 & \sqrt{3}+1 \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}, n=1,2,3$ and

$a_1 = b_1 = 1$ then $a_{2010} =$

- A) 2^{3013} B) 3^{3013} C) 2^{3015} D) 3^{3015}

5. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$

$Q = PAP^T$ and if $P^T Q^{2012} P = \begin{bmatrix} \alpha & \gamma \\ \beta & \alpha \end{bmatrix}$, then

the value of $(\alpha + \beta)$ is

- A) 1 B) -1 C) 2 D) -2

6. A and B are two non-singular matrices such that $A^6 = I$ and $AB^2 = BA (B \neq I)$. A value of k so that $B^k = I$ is

- A) 31 B) 32 C) 64 D) 63

7. If A is a 2×2 matrix such that $A^2 - 4A + 3I = 0$ then the inverse of $A+3I$ is

- A) $\frac{1}{24}A - \frac{7}{24}I$ B) $\frac{1}{21}A - \frac{7}{21}I$

- C) $\frac{7}{24}I - \frac{1}{24}A$ D) $A-3I$

8. If $A = [a_{ij}]_{4 \times 4}$ such that

$$a_{ij} = \begin{cases} 2 & \text{when } i = j \\ 0 & \text{when } i \neq j \end{cases} \text{ then}$$

$$\left\{ \frac{\det(\text{adj}(\text{adj} A))}{7} \right\} \text{ is (where } \{ \} \text{ represents fractional part of } x) \text{ is}$$

- A) $\frac{1}{7}$ B) $\frac{2}{7}$ C) $\frac{3}{7}$ D) $\frac{9}{7}$

9. If A, B, C are three square matrices of same order satisfying $A^2 = A^{-1}$ and let

$B = A^{2^n}$ and $C = A^{2^{n-2}} (n \in N)$ then

$\det(B-C) =$

- A) 0 B) 1 C) -1 D) $(-1)^n$

10. If A is an idempotent matrix satisfying $(I - 0.4A)^{-1} = I - \alpha A$ then $\alpha =$

- A) $-\frac{2}{3}$ B) $\frac{1}{3}$ C) $\frac{1}{9}$ D) $\frac{1}{8}$

11. If A is a 3×3 matrix such that $|A| = 4$ then

$$|(\text{adj} A)^{-1}| =$$

- A) 16 B) 64 C) $\frac{1}{64}$ D) $\frac{1}{16}$

12. If A is 3×3 non-singular matrix then

$$\det(A^{-1} \text{adj} A) =$$

- A) $\det A$ B) 1
C) $(\det A)^2$ D) $(\det A)^{-1}$

13. Let M and N be two 3×3 non-singular

symmetric matrices such that $MN = NM$.

If P^T denotes the transpose of the matrix P, then $M^2 N^2 (M^T N)^{-1} (MN^{-1})^T$ is equal to

- A) M^2 B) N^2 C) MN D) NM

14. If $A \neq A^2 = I$ then $\det(I+A) =$

- A) 0 B) 1 C) -1 D) 2

15. Let A be an n^{th} - order square matrix and B be adjoint of A then $|AB + KI_n|$ is (where K is scalar quantity)

- A) $(|A| + K)^{n-2}$ B) $(|A| + K)^n$
C) $(|A| + K)^{n-1}$ D) None of these

16. Let $A = \begin{pmatrix} -5 & -8 & -7 \\ 3 & 5 & 4 \\ 2 & 3 & 3 \end{pmatrix}$, $B = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. If AB is a

scalar ($\neq 0$) multiple of B, then $x + y =$

- A) z B) -z C) 0 D) 2z

17. let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $A^3 = 0$ then the sum

of all the elements of A^2

- A) 0 B) 1 C) 2 D) 3

18. If $\lambda_1, \lambda_2, \lambda_3$ are the direction cosines of a line

then the matrix $\begin{bmatrix} \lambda_1^2 & \lambda_1 \lambda_2 & \lambda_1 \lambda_3 \\ \lambda_1 \lambda_2 & \lambda_2^2 & \lambda_2 \lambda_3 \\ \lambda_1 \lambda_3 & \lambda_2 \lambda_3 & \lambda_3^2 \end{bmatrix}$ is

- A) Idempotent B) Involuntary
C) Nilpotent D) Orthogonal

19. The value of λ such that the system $x - 2y + z = -4$, $2x - y + 2z = 2$, $x + y + \lambda z = 4$ has no solution is

- A) 0 B) 1 C) -1 D) 3

20. The number of values of k for which the system of equations $(k+1)x + 8y = 4k$, $kx + (k+3)y = 3k-1$ has infinite number of solutions is

- A) 1 B) 2 C) 3 D) 4

21. If $x = \frac{a}{b-c}$, $y = \frac{b}{c-a}$, $z = \frac{c}{a-b}$ where

$a \neq b \neq c \neq 0$ then $xy + yz + zx =$

- A) 0 B) -1 C) 1 D) 2

22. The ordered triplet (s) (x, y, z) of real numbers satisfying

$$5\left(x + \frac{1}{x}\right) = 12\left(y + \frac{1}{y}\right) = 13\left(z + \frac{1}{z}\right)$$

and $xy + yz + zx = 1$ is given by

- A) $\left(\frac{1}{5}, \frac{2}{3}, 1\right)$ B) $\left(1, \frac{2}{3}, \frac{5}{3}\right)$
C) $\left(-\frac{1}{5}, -\frac{2}{3}, -1\right)$ D) $\left(-\frac{1}{5}, -1, -\frac{2}{3}\right)$

23. If t is real and $\lambda = \frac{t^2 - 3t + 4}{t^2 + 3t + 4}$ then the number of solutions of system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$ and $6x + 5y + \lambda z = -3$ is

- A) One B) Two C) Zero D) Four

MULTI ANSWER QUESTIONS

24. Let $A = a_{ij}$ be a matrix of order 3×3

where $a_{ij} = \begin{cases} x, & \text{if } i = j, x \in R \\ 1, & \text{if } |i - j| = 1 \\ 0, & \text{otherwise} \end{cases}$ then which of

the following statement(s) is/are true

- a) for $x = 2$, A is a diagonal matrix
b) A is symmetrical matrix
c) for $x = 2$, determinant of A is 6
d) Let $f(x) = \text{determinant of A}$, then $f(x)$ has both local minimum and maximum.

25. If a square matrix $A = [a_{ij}]$, $a_{ij} = \frac{i^2 - j^2}{4}$ is of even order then

- A) A is a skew - symmetric
B) $|A|$ is a perfect square
C) A is symmetric and $|A| = 0$
D) A is neither symmetric nor skew symmetric

26. Let a, b, c, d be four real numbers satisfying the system of equation $a+b=8$, $ab+c+d=23$,

$ad+bc=28$ and $cd=12$. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then

the trace of A is

- A) 8 B) 9 C) 7 D) 6

27. Let $[A_k]_{n \times n}$ be a square matrix of order $n \times n$,

such that $a_{ij} = \begin{cases} 0, i \neq j \\ \frac{1}{k+j}, i = j \end{cases}$ and $[B_k]_{n \times n}$ is its

inverse matrix, then which is/are true?

A) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B_k)_{n \times n}}{m^3} \right) = \frac{1}{6}$

B) $\sum_{n=1}^{10} \text{trace}(B_2)_{n \times n} = 320$

C) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B_k)_{n \times n}}{m^3} \right) = \frac{1}{3}$

D) $\sum_{n=1}^{10} \text{trace}(B_2)_{n \times n} = 330$

28. Let A be the 2×2 matrix given by $A = (a_{ij})$

where $a_{ij} \in \{0, 1, 2, 3, 4\}$ such that

$a_{11} + a_{12} + a_{21} + a_{22} = 4$ then which of the following statement(s) is/are true?

- A) Number of matrices A such that the trace of A is equal to 4, is 5
 B) Number of matrices A, such that A is invertible is 18
 C) Absolute difference between maximum value and minimum value of $\det(A)$ is 8
 D) Number of matrices A such that A is either symmetric (or) skew symmetric and $\det(A)$ is divisible by 2, is 5

29. Which of the following is/are true?

A) If A is a square matrix such that $A^2 = A$ then $(I + A)^3 - 7A = I$

B) If A is a square matrix such that $A^2 = A$ then

$$(I + A)^3 - 7A = O$$

C) Let B and C be two square matrices such that $BC = CB$ and $C^2 = O$. If $A = B + C$ then

$$A^3 - B^3 - 3B^2C = O$$

D) Let B and C be two square matrices such that $BC = CB$ and $C^2 = O$. If $A = B + C$ then

$$A^3 + B^3 - 3B^2C = O$$

30. If a, b, c are the roots of $x^3 + x^2 - 4 = 0$ and

$$A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix} \text{ then}$$

- A) A is non singular B) A is orthogonal

C) $AA^T = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$ D) $A^{-1} = A^T$

31. Which of the following statements is/are true

A) If A is a skew symmetric matrix of order $n \times n$, then the maximum number of non-zero elements in A, is $n(n-1)$

B) If A is a symmetric matrix of order $n \times n$, then maximum number of non-zero elements is A, is n^2

C) Minimum number of zero elements in an upper triangular matrix of order $n \times n$ is $\frac{n(n-1)}{2}$

D) total number of matrices that can be formed using all 5 different letters such that no letter is repeated in any matrix is $2.5!$

32. The rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is

- A) 2 if $a = 6$ B) 2 if $a = 1$
 C) 1 if $a = 2$ D) 1 if $a = -6$

33. If the rank of $A = \begin{bmatrix} y+a & b & c \\ a & y+b & c \\ a & b & y+c \end{bmatrix}$ is

3 then y cannot be equal to

MATRICES

- A) $a+b+c$ B) $-(a+b+c)$
C) 0 D) 1
34. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $aA^{-1} = bA^2 + cA + dI$ where G.C.D of a,d is one then
A) $a=6$ B) $b=1$ C) $c=-6$ D) $d=1$
35. Let A_n is a $n \times n$ matrix in which diagonal elements are $1, 2, 3, \dots, n$ (i.e., $a_{11}=1, a_{22}=2, a_{33}=3, \dots, a_{ii}=i, \dots, a_{nn}=n$) and all other elements are equal to 'n' then
A) A_n is not invertible for all n
B) A_n is invertible for all n
C) $\det A_5 = 120$
D) $\det A_n = 0$
36. If the matrix A and B are of 3×3 and $(I - AB)$ is invertible then which of the following statements is/are correct?
A) $I - BA$ is not invertible
B) $I - BA$ is invertible
C) $I - BA$ has for its inverse $I + B(I - AB)^{-1}A$
D) $I - BA$ has for its inverse $I + A(I - BA)^{-1}B$
37. If $\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ then
A) $a = \cos 2\theta$ B) $a = 1$
C) $b = \sin 2\theta$ D) $b = 1$
38. If the system $(a-t)x + by + cz = 0$, $bx + (c-t)y + az = 0$, $cx + ay + (b-t)z = 0$ has non trivial solution then
A) $t = a+b+c$
B) $t^2 = a^2 + b^2 + c^2 - ab - bc - ca$
C) $t^2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ D) $t^3 = \begin{vmatrix} a & b & c \\ c & a & b \\ b & a & c \end{vmatrix}$

JEE ADVANCED - VOL - I

39. If $a > b > c$ and the system of equations and $cx + ay + bz = 0$ has a non trivial solution then both the roots of equation $at^2 + bt + c = 0$ are
A) real B) of opposite sign
C) positive D) complex

COMPREHENSION QUESTIONS

Passage - I

A Pythagorean triplet is triplet of positive integers (a,b,c) such that $a^2 + b^2 = c^2$. Define matrices P, Q, R by

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}, Q = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -2 \\ 2 & 2 & 3 \end{bmatrix} \text{ and } R = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

40. If we write Pythagorean triplet (a,b,c) in matrix form as $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ then which of the following matrix product is not Pythagorean triplet?

A) $Q \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ B) $P \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ C) $R \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ D) None

41. Which one of the following does not hold good?
A) $P^{-1} = \text{adj}P$ B) $(PQ)^{-1} = \text{adj}(PQ)$
C) $(QR)^{-1} = \text{adj}(QR)$
D) $(PQR)^{-1} \neq \text{adj}(PQR)$
42. Trace of $(P + Q^T + 2R) =$
A) 17 B) 15 C) 14 D) 18

Passage - II

A matrix A is in echelon form if

- Every row of A has all its elements zero, occurs below a row which has non zero elements.
- The first non zero element in each non

zero row = 1

3) Number of zeros before non zero element in each row is less than no of zeros in the next row Now number of non zero rows in the echelon form on matrix is the rank of the matrix

43. Rank of the matrix $\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 5 & 2 \end{bmatrix}$

A) 1 B) 2 C) 3 D) 4

44. If the rank of the matrix $\begin{bmatrix} 4 & 2 & 1-x \\ 5 & k & 1 \\ 6 & 3 & 1+x \end{bmatrix}$ is 2

then

A) $k = \frac{5}{2}, x = \frac{1}{5}$ B) $k = \frac{5}{2}, x \neq \frac{1}{5}$

C) $k = \frac{1}{5}, x = \frac{5}{2}$ D) $k \neq \frac{1}{5}, x \neq \frac{5}{2}$

45. Suppose λ is real and

$A = \begin{bmatrix} 1 & 4 & 5 \\ \lambda & 8 & 8\lambda - 6 \\ \lambda^2 + 1 & 8\lambda + 4 & 2\lambda + 21 \end{bmatrix}$ then

A) $\lambda = 2 \Rightarrow \text{Rank of } A = 1$

B) $\lambda \neq 2 \Rightarrow \text{Rank of } A = 3$

C) $\lambda = 1 \Rightarrow \text{Rank of } A = 2$

D) $\lambda \neq -1 \Rightarrow \text{Rank of } A = 3$

Passage - III

A is a square matrix, said to be a idempotent if $A^2 = A$. Nilpotent if $A^k = 0$, $k \in \mathbb{N}$. Involutory if $A^2 = I$

$A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix},$

$B = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}, C = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$

46. Which is nilpotent?

A) A B) B C) C D) AC^2

47. Which is not idempotent?

A) A^3C^2 B) A^2C^2 C) BC^2 D) C^2A

48. Which has multiplicative inverse?

A) BC^2 B) A^3C^2 C) A^2B D) C^3

Passage - IV

Let A be a $m \times n$ matrix, if there exists a matrix "L" of type $n \times m$ such that $LA = I_n$, then "L" is called left inverse of A. Similarly if there exists a matrix 'R' of type $n \times m$ such that $AR = I_m$ then 'R' is called right inverse of A.

49. Which of the following is not left inverse of

$\begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}$

A) $\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$

B) $\begin{pmatrix} 2 & -7 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$

C) $\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$

D) $\begin{pmatrix} 0 & 3 & -1 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$

50. Number of right inverse matrix of the matrix

$\begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}$ is

A) 0

B) 1

C) 2

D) infinitely many

51. For which of the following matrices number of left inverses is greater than the number of right inverses

A) $\begin{pmatrix} 1 & 2 & 4 \\ -3 & 2 & 1 \end{pmatrix}$

B) $\begin{pmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{pmatrix}$

C) $\begin{pmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{pmatrix}$

D) $\begin{pmatrix} 3 & 3 \\ 1 & 1 \\ 4 & 4 \end{pmatrix}$

Passage - V

Let a, b and c be three real numbers satisfying

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \dots\dots\dots(\text{E})$$

(IIT 2011)

52. If the point $P(a, b, c)$, with reference to E lies on the plane $2x + y + z = 1$ then the value of $7a + b + c$ is

- A) 0 B) 12 C) 7 D) 6

53. Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with b and c satisfying

E, then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to

- A) -2 B) 2 C) 3 D) -3

54. Let $b = 6$, with a and c satisfying E. If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then

$$\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n \text{ is}$$

- A) 6 B) 7 C) 6/7 D) ∞

Passage - VI

A 3×3 determinant has entries either 1 or -1.

Let $S_3 =$ set of all determinants which contain product of elements of any row or any column is -1

For example

$$\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} \text{ is an element of the set } S_3.$$

55. Number of elements of the set $S_3 =$

- A) 10 B) 16 C) 12 D) 18

56. Number of elements of the set $S_n =$

- A) 2^n B) 2^{n-1} C) 2^{2n} D) $2^{(n-1)^2}$

57. The minimum value out of all elements of the set S_3 is

- A) -4 B) -6 C) 4 D) 6

58. Column-I

A) A is a square matrix such that $A^2 = A$.

If $(I + A)^8 = I + \lambda A$ then $\lambda + 1$ is

B) If A is a square matrix of order 3 such that

$|A| = 2$ then $\left| (adJA^{-1})^{-1} \right|$ is

C) Let $|A| = |a_{ij}|_{3 \times 3} \neq 0$ Each element a_{ij} is

multiplied by λ^{i-j} . Let $|B|$ the resulting determinant where $|A| = k|B|$ then k is

D) If A is a diagonal matrix of order 3×3 is commutative with every square matrix of order 3×3 under multiplication and trace

$(A) = 12$, then $|A| =$

Column-II

P) 64

Q) 1

R) 256

S) 4

59. Match the following:

The matrix $A = [a_{ij}]_{2 \times 2} = \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}$ and

$S = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ are such that

$S^T A S =$ diagonal matrix $= D = [d_{ij}]_{2 \times 2}$ for

some suitably chosen value of $\theta \in \left(0, \frac{\pi}{4} \right)$ Let

$$M = A^{50} = [m_{ij}]_{2 \times 2}$$

Column - I

A) If θ is the angle that makes $S^T A S$ as a diagonal matrix then $3 \tan 2\theta$ is

B) The value of $\left| \frac{4m_{11} - m_{22}}{3^{51}} \right|$ is

C) The value of $d_{11} - d_{22}$ is

MATRIX MATCHING QUESTIONS

D) The value of $\left| \frac{m_{12} - m_{21}}{3^{50}} \right|$ is

Column - II

- p) 0
q) 4
r) 1
s) 5

60. The entries of 3×3 matrix either 1 or -1

List - I

- A) Total number of symmetric matrices
B) Maximum value of the determinants so formed
C) The maximum value of the trace of the matrix
D) No of skew symmetric matrices

List - II

- p) 0
q) 3
r) 4
s) 64
t) 32

INTEGER QUESTIONS

61. The number of all possible value of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y+z)\cos 3\theta = xyz \sin 3\theta,$$

$$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z},$$

$(xyz) \sin 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$ have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

(IIT - 2010)

62. If the system of equations

$\lambda x + y + z = 0$, $-x + \lambda y + z = 0$, $-x - y + \lambda z = 0$ will have a non-zero solution then the real values of λ is

(IIT- 1982)

63. If $a_k = k(10_{C_k})$, $b_k = (10-k)(10_{C_k})$ and

$$A_k = \begin{pmatrix} a_k & 0 \\ 0 & b_k \end{pmatrix} \text{ and } A = \sum_{k=1}^9 A_k = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

then the sum of digits in $\frac{ab}{2^9 - 1}$ is

64. $A \neq I$, $B^3 = I$, $AB = BA^2$ and $A^k = I$ then the least value of k is

65. Let $A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$ is able to express as

$B^3 + C^3$ where B, C are two matrices then

the value of $|\text{trace of B} + \text{trace of C}| =$

66. If $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ and A is an orthogonal

matrix then the product of a and b is equal to

67. If the product of n matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \text{ is equal to}$$

the matrix $\begin{pmatrix} 1 & 378 \\ 0 & 1 \end{pmatrix}$ and $n = a^3$ (where $a \in \mathbb{N}$) then a =

68. If $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and $A^n = A^{n-2} + A^2 - I$ and

$$A^{50} = kA^2 - 24I \text{ then } \left[\frac{k}{6} \right] \text{ (where } [\cdot] \text{ G.I.F.)} =$$

69. If $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\det(A^n - I) = 1 - \lambda^n$, $n \in \mathbb{N}$ then $\lambda =$

70. The values of k (say k_1, k_2 and $k_1 < k_2$) for which the planes $kx + 4y + z = 0$, $2x + 2y + z = 0$ and $4x + ky + 2z = 0$

intersect in a line and $P = \begin{pmatrix} k_2 & k_1 \\ k_1 & k_2 \end{pmatrix}$ then

$$\frac{1}{24} \det(P^{-1}) = \dots$$

MATRICES

71. Let $a_k = {}^nC_k$ ($0 \leq k \leq n$) and $A_k = \begin{pmatrix} a_{k-1} & 0 \\ 0 & a_k \end{pmatrix}$

$$B = \sum_{k=1}^{n-1} A_k \cdot A_{k+1} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \text{ then } \frac{a}{b} =$$

72. Let $A = \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ and

$$C_r = \begin{pmatrix} r-3^r & 2^r \\ 0 & (r-1)3^r \end{pmatrix} \text{ be given 3 matrices}$$

then the value of $\sum_{r=1}^{50} \text{tr}((AB)^r C_r) = 3^b a + 3$

$$\text{then } \frac{a+b}{25} =$$

73. Let α, β ($\alpha < \beta$) be the solutions of

$$\sum_{n=1}^6 \frac{1}{\sin\left(\theta + (n-1)\frac{\pi}{4}\right) \sin\left(\theta + \frac{n\pi}{4}\right)} = 4\sqrt{2} \quad \text{and}$$

determinant value of A where

$$A = \begin{pmatrix} \sin \beta & \sin \alpha \\ \sin \alpha & \sin \beta \end{pmatrix} \text{ is k then } 8k^2 =$$

SUBJECTIVE QUESTIONS

74. Find all the values of t for which the system of equations

$$(t-1)x + (3t+1)y + 2tz = 0,$$

$$(t-1)x + (4t-2)y + (t+3)z = 0 \text{ and}$$

$$2x + (3t+1)y + 3(t-1)z = 0$$

has a non-trivial solution.

75. If the system of equations $x + ky + 3z = 0$, $3x + ky - 2z = 0$, $2x + 3y - 4z = 0$ possess a non trivial solution over the set of

rational number then $\left| \frac{z}{y} \right|$ is

76. A, B, C are 2x2 matrices with real entries

$$\text{given } A^*B = \frac{AB^T + A^TB}{2}.$$

Then prove that

JEE ADVANCED - VOL - I

$$\text{i) } A^*B = B^*A$$

$$\text{ii) } A^*(B+C) = A^*B + A^*C$$

77. Let A, B two matrices such that $AB=BA$ then prove that for $n \in N$

$$(A^n - B^n)(A^n + B^n) = A^{2n} - B^{2n}$$

KEY LEVEL -VI

SINGLE ANSWER QUESTIONS

1.C	2.A	3.C	4.A
5.A	6.D	7.C	8.A
9.A	10.A	11.D	12.A
13.A	14.A	15.B	16.B
17.A	18.A	19.B	20.A
21.B	22.A	23.A	

MULTI ANSWER QUESTIONS

24.ABCD	25.AB	26.ABC	27.AD
28.ABCD	29.AC	30.ABD	31.ABCD
32.ABD	33.BC	34.ABC	35.BC
36.BC	37.AC	38.AB	39.ACD

COMPREHENSION QUESTIONS

40.B	41.D	42.C	43.B
44.B	45.A	46.B	47.C
48.D	49.C	50.A	51.C
52.A	53.A	54.B	55.B
56.D	57.A		

MATRIX MATCHING QUESTIONS

58. A-R, B-S; C-Q, D-P

59. A-Q, B-R, C-S, D-P

60. A-S, B-R, C-Q, D-P

INTEGER QUESTIONS

61.3	62.0	63.7	64.7
65.3	66.2	67.3	68.4
69.2	70.2	71.1	72.4
73.6	74.0(or)3	75.3	

HINTS LEVEL -VI

SINGLE ANSWER QUESTIONS

- The first element in $m(17)$ is 1497 and difference of diagonal elements is 17 Trace of

$$m(17) = \frac{17}{2} [2 \times 1497 + 16 \times 17] = 27,761$$

2. Characteristic equation is

$$\lambda^3 - 5\lambda^2 + \lambda + 3 = 0$$

$$\text{Trace of } A^3 = \lambda_1^3 + \lambda_2^3 + \lambda_3^3$$

$$= (\sum \lambda_i) [(\sum \lambda_i)^2 - 3 \sum \lambda_i \lambda_j] + 3 \lambda_1 \lambda_2 \lambda_3$$

$$= 101$$

3. Given that $X = AB + BA \Rightarrow X = X^T$ and

$$y = AB - BA \Rightarrow y = -y^T$$

$$\text{Now, } (xy)^T = y^T x^T = -YX.$$

$$4. \begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = 2\sqrt{2} \begin{pmatrix} \cos \frac{\pi}{12} & -\sin \frac{\pi}{12} \\ \sin \frac{\pi}{12} & \cos \frac{\pi}{12} \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}$$

$$\begin{pmatrix} a_{2010} \\ b_{2010} \end{pmatrix} = (2\sqrt{2})^{2009} \begin{pmatrix} \cos \frac{2009\pi}{12} & -\sin \frac{2009\pi}{12} \\ \sin \frac{2009\pi}{12} & \cos \frac{2009\pi}{12} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$a_{2010} = (2\sqrt{2})^{2009} \left(\cos \left(\frac{2009\pi}{12} \right) - \sin \left(\frac{2009\pi}{12} \right) \right)$$

$$= (2\sqrt{2})^{2009} \cdot \frac{1}{2} = 2^{3013}$$

$$5. Q^{2012} = (PAP^T)(PAP^T) \dots (PAP^T) (2012 \text{ times})$$

$$= PAAP^T \dots = PA^{2012}P^T$$

$$P^T P A^{2012} P^T P = A^{2012} \text{ But}$$

$$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\therefore A^{2012} = \begin{bmatrix} 1 & 2012 \\ 0 & 1 \end{bmatrix}$$

$$\therefore \alpha = 1, \beta = 0$$

$$6. A^6 = I \Rightarrow BA^6 = B \Rightarrow (BA)A^5 = B$$

$$\Rightarrow AB^2 A^5 = B$$

$$\Rightarrow AB(AB^2)A^4 = B$$

$$\Rightarrow A^2 B^4 A^4 = B$$

Proceeding like this we get

$$A^6 B^{64} = B \Rightarrow B^{64} = B$$

$$\Rightarrow B^{63} = I$$

$$\Rightarrow K = 63$$

$$7. A^2 - 4A + 3I = 0 \Rightarrow A(A + 3I) - 7A + 3I = 0$$

$$\Rightarrow A(A + 3I) - 7(A + 3I) = -24I$$

$$\Rightarrow (A + 3I)(A - 7I) = -24I$$

$$\Rightarrow (A + 3I) \left(\frac{7}{24}I - \frac{A}{24} \right) = I$$

$$\Rightarrow (A + 3I)^{-1} = \frac{7}{24}I - \frac{A}{24}$$

$$8. |A| = 2^4 = 16 \Rightarrow |adj(adJA)| = 16^9$$

$$\therefore \left\{ \frac{16^9}{7} \right\} = \left\{ \frac{2^9(1+7)^9}{7} \right\} = \left\{ \frac{2^9}{7} \right\} = \left\{ \frac{8^3}{7} \right\} = \frac{1}{7}$$

$$9. B = A^{2^n} = A^{2 \cdot 2^{n-1}} = (A^{-1})^{2^{n-1}}$$

$$= (A^{-1})^{2 \cdot 2^{n-2}} = ((A^{-1})^2)^{2^{n-2}}$$

$$= ((A^2)^{-1})^{2^{n-2}} = A^{2^{n-2}}$$

$$= C$$

$$\Rightarrow B - C = 0 \Rightarrow |B - C| = 0$$

$$10. (I - 0.4A)(I - \alpha A) = I$$

$$11. |(adj A)^{-1}| = \frac{1}{|adj A|}$$

$$12. \text{Use Det}(AB) = \text{Det } A \cdot \text{Det } B$$

$$13. M^2 N^2 N^{-1} (M^{-1})^T (N^{-1})^T M^T$$

$$= M^2 N^2 N^{-1} M^{-1} N^{-1} M$$

$$= M^2$$

$$14. A^2 - I = 0 \Rightarrow (A + I)(A - I) = 0$$

$$\det |A + I| \neq 0 \Rightarrow A - I = 0$$

$$\Rightarrow A = I$$

$$\therefore \det(A + I) = 0$$

$$15. \text{We have, } AB = A(adj A) = |A|I_n$$

$$\therefore AB + kI_n = |A|I_n + kI_n$$

$$\Rightarrow AB + KI_n = (|A| + k)I_n$$

$$\Rightarrow |AB + kI_n| = (|A| + k)|I_n|$$

$$(\because |\alpha I_n| = \alpha^n)$$

MATRICES

16. $AB = \lambda B \Rightarrow -5x - 8y - 14 = \lambda x$
 $3x + 5y + 8 = \lambda y$
 $2x + 3y + 6 = 2\lambda$
 adding above equations
 $\lambda(x + y + z) = 0$
 $\Rightarrow (x + y) = -z$
17. $A^3 = 0 \Rightarrow |A| = 0$ by characteristic equation
 $A^2 - (a + d)A = 0$. $A^3 - (a + d)A^2 = 0$
 $\Rightarrow (a + d)A^2 = 0$
 if $(a + d) \neq 0$ then $A^2 = 0$.
 $(a + d) = 0$ then also $A^2 = 0$
18. $A^2 = A \Rightarrow$ it is idempotent
19. $\begin{vmatrix} 1 & -2 & 1 \\ 2 & -1 & 2 \\ 1 & 1 & \lambda \end{vmatrix} = 0$
20. $\frac{K+1}{K} = \frac{8}{K+3} = \frac{4K}{3K-1} \Rightarrow K^2 + 4K + 3 = 8K$
 $\Rightarrow K^2 - 4K + 3 = 0 \Rightarrow K = 1, 3$
21. $\begin{vmatrix} 1 & -x & x \\ y & 1 & -y \\ z & -z & -1 \end{vmatrix} = 0$
22. Let $x = \tan \alpha$, $y = \tan \beta$, $z = \tan \gamma$
 $\Rightarrow \sum \tan \alpha \tan \beta = 1 \Rightarrow \alpha + \beta + \gamma$
 $= (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$
 $\therefore \frac{5}{\sin 2\alpha} = \frac{12}{\sin 2\beta} = \frac{13}{\sin 2\gamma}$
 $\Rightarrow \sin^2 2\alpha + \sin^2 2\beta = \sin^2 2\gamma$
 $\alpha + \beta = \gamma \Rightarrow 2\gamma = 90^\circ \Rightarrow \gamma = 45^\circ \Rightarrow \alpha + \beta = 45^\circ$
 $\Rightarrow z = 1 \Rightarrow x + \frac{1}{x} = \frac{26}{5} \Rightarrow x = \frac{1}{5}$
 $y + \frac{1}{y} = \frac{13}{6} \Rightarrow y = \frac{2}{3}$
 $\therefore (x, y, z) = \left(\frac{1}{5}, \frac{2}{3}, 1\right)$
23. $f(\pm 2) = \frac{4-6+4}{4+6+4} = \frac{1}{7} \Rightarrow x \in \left(\frac{1}{7}, 7\right)$

$$\begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix}$$

$$= 3(2\lambda + 15) + 1(\lambda + 18) + 4(-7) = 7\lambda + 35$$

For $\lambda \in \left(\frac{1}{7}, 7\right)$, $7\lambda + 35 \neq 0 \Rightarrow$ the system has unique solution

MULTI ANSWER QUESTIONS

24. $A = \begin{pmatrix} x & 1 & 0 \\ 1 & x & 1 \\ 0 & 1 & x \end{pmatrix}$ and $|A| = x^3 - 2x$
25. For simplicity we just consider a fourth order matrix, then
 $A = \begin{bmatrix} 0 & -1 & -8 & -15 \\ 1 & 0 & -5 & -12 \\ 8 & 5 & 0 & -7 \\ 15 & 12 & 7 & 0 \end{bmatrix}$
 Which is a skew-symmetric. Also,
 $|A| = (1 \times 5 - 8 \times 12 \times 15)^2 = \text{perfect square.}$
26. $(x^2 + ax + c)(x^2 + bx + d)$
 $= x^4 + (a+b)x^3 + (ab+c+d)x^2 + (ad+bc)x + cd$
 $= x^4 + 8x^3 + 23x^2 + 28x + 12$
 $= (x+1)(x+2)^2(x+3)$
27. $[B_k]_{n \times n}$ is a matrix, such that
 $b_{ij} = 0, i \neq j$ and $b_{ii} = k + i$
 $\text{trace}(B_k)_{n \times n} = \sum_{i=1}^n (k+i) = k_n + \frac{n(n+1)}{2}$
 $\sum_{n=1}^m \text{trace}(B_k)_{n \times n} = \frac{km(m+1)}{2} + \frac{m(m+1)(m+2)}{6}$
 $\sum_{n=1}^{10} \text{trace}(B_2)_{n \times n} = \frac{2 \times 10 \times 11}{2} + \frac{10 \times 11 \times 12}{6} = 330$
 $\lim_{m \rightarrow \infty} \left(\frac{\sum_{n=1}^m \text{trace}(B_k)_{n \times n}}{m^3} \right) = \frac{1}{6}$
28. a) $\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$
 b) using 0, 0, 2, 2 $\rightarrow$ there are two matrices which

are invertible $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$

using 0,0,1,3 $\rightarrow$ there are four matrices which are invertible

using 0,1,1,2 $\rightarrow$ there are twelve matrices which are invertible

using 0,0,0,4 and using 1,1,1,1 no matrix is formed, which is invertible

$\therefore$ total 18

c) $|4 - (-4)| = 8$

d) there are five matrices, which are either symmetric or skew symmetric and whose determinant is divisible by 2

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

29. $(I + A)^3 = I^3 + 3A + 3A^2 + A^3 = I + 7A$

$$A = B + C \Rightarrow A^3 = (B + C)^3$$

$$= B^3 + 3B^2C + 3BC^2 + C^3$$

$$\Rightarrow A^3 - B^3 - 3B^2C = 3BC^2 + C^3 = 0$$

30. $a + b + c = -1$, $ab + bc + ca = 0$, $abc = 4$

$$\det A = a^3 + b^3 + c^3 - 3abc$$

$$= -1(1) = -1 \text{ and } AA^T = I$$

31. abcd

32. Echelon form is

$$\begin{bmatrix} -1 & 2 & 5 \\ 0 & 0 & a+6 \\ 0 & 0 & a+6 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & 5 \\ 0 & 0 & a+6 \\ 0 & 0 & 0 \end{bmatrix}$$

33. $(y+a)((y+b)(y+c)-bc)$
 $-b(ay+c(ab-a(y+b))) \neq 0$

34. Given equation $|A - \lambda I| = 0$

$$\lambda^3 - 6\lambda^2 + 11\lambda = 6$$

$$\therefore A^2 - 6A + 11I = 6A^{-1}$$

35. $\det A_5 = 1.2.3.4.5 = 120$

36. Let $(I - AB)^{-1} = P$

$$\Rightarrow P(I - AB) = I \Rightarrow P - PAB = I$$

$$PB^{-1} - PA = B^{-1}$$

$$BPB^{-1} - BPA = I$$

$$BPB^{-1} = I + BPA$$

$$\text{now } BPB^{-1} = B(I - AB)^{-1}B^{-1}$$

$$= B(B(I - AB))^{-1}$$

$$= (B^{-1})^{-1}(B(I - AB))^{-1} = (B(I - AB)B^{-1})^{-1}$$

$$= ((B - BAB)B^{-1})^{-1}$$

$$= (I - BA)^{-1}$$

37. $\begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{bmatrix}^{-1}$
 $= \begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \left(\frac{1}{1 + \tan^2 \theta} \right) \begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix}$

$$= \frac{1}{1 + \tan^2 \theta} \begin{bmatrix} 1 - \tan^2 \theta & -2 \tan \theta \\ 2 \tan \theta & 1 - \tan^2 \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$\therefore a = \cos 2\theta, b = \sin 2\theta$$

38. $\begin{vmatrix} a-t & b & c \\ b & c-t & a \\ c & a & b-t \end{vmatrix} = 0$

39. $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$$\Rightarrow a + b + c = 0 \text{ or } a = b = c$$

$$a > b > c \Rightarrow a + b + c = 0$$

$$\Rightarrow 1 \text{ is a root of } at^2 + bt + c = 0$$

$$\text{And other root is } \frac{c}{a} > 0$$

COMPREHENSION QUESTIONS

40,41,42

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 26 \\ 20 \\ 29 \end{bmatrix}$$

43,44,45

$$\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 5 & 2 \end{bmatrix} \quad \text{echelon form is}$$

$$\begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Det value} \Rightarrow (5 - 2K)(1 - 5\lambda) = 0$$

$$\text{Is } K = \frac{5}{2} \text{ or } x = \frac{1}{5}$$

$$\text{Det value} = (\lambda - 2)^2(\lambda + 1) = 0$$

$$\lambda = -1 \Rightarrow \text{Rank} = 2$$

$$\lambda = 2 \Rightarrow \text{Rank} = 1$$

46,47,48

we observe that $A^2 = A$, $B^2 = 0$, $C^2 = I$ and $|A| = 0, |B| = 0$ and $|C| = -1$ and hence results, required

49,50,51

i) option (c) is not the left inverse of $\begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}$

$$\text{ii) } \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow a - d = 1, 2a + 3d = 0, a + d = 0 \&$$

$$b - e = 0, b + e = 1, 2b + 3e = 0$$

$$c - f = 0, c + f = 0, 2c + 3f = 1$$

which are not consistent

iii) option (c) has more left inverse, then the number of right inverse

52,53,54

Conceptual

55,56,57

$$2a + b + c = 1, \text{ we have}$$

$$-\frac{2k}{7} - \frac{6k}{7} + k = 1 \Rightarrow k = -7$$

$$\text{Thus } 7a + b + c = 7 + 6 - 7 = 6.$$

$$(20) \text{ For } a = 2, k \text{ turn to be } k = -14$$

$$\therefore b = 12, c = -14$$

$$\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c} = \frac{3}{\omega^2} + \frac{1}{\omega^{12}} + 3 \cdot \omega^{14}$$

$$= 3\omega + 1 + 3\omega^2 = 3(\omega + \omega^2) + 1 = -3 + 1 = -2.$$

$$(21) b = 6 \Rightarrow k = -7$$

$$\therefore a = 1 \text{ and } c = -7$$

So the equation become

$$x^2 + 6x - 7 = 0 \therefore x = -7, 1$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n = \sum_{n=0}^{\infty} \left(\frac{6}{7} \right)^n = \frac{1}{1 - (6/7)} = 7.$$

MATRIX MATCHING QUESTIONS

$$\begin{aligned} 58. \quad A^2 = A &\Rightarrow (I + A)^2 = (I + A)(I + A) \\ &= I + 3A \Rightarrow (I + A)^8 = I + (2^8 - 1)A \\ &\Rightarrow \lambda + 1 = 2^8 \end{aligned}$$

$$\text{B) } \frac{1}{|\text{adj}(A^{-1})|} = \frac{1}{|A^{-1}|^2} = |A|^2 = 4$$

59. Conceptual

60. Conceptual

INTEGER QUESTIONS

$$\begin{aligned} 61. \quad &\text{from the equations we get} \\ &y \cos 3\theta - y \sin 3\theta = 0 \\ &\Rightarrow \tan 3\theta = 1 (y \neq 0) \Rightarrow \pi/4 + n\pi = 3\theta \\ &\Rightarrow \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12} \text{ are the solutions.} \end{aligned}$$

$$62. \quad \begin{vmatrix} \lambda & 1 & 1 \\ -1 & \lambda & 1 \\ -1 & -1 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 0$$

$$63. \quad a = \sum_{k=1}^9 a_k = \sum_{k=1}^9 k \cdot 10_{C_k} = 10 \sum_{k=1}^9 9_{C_{k-1}} = 10(2^9 - 1)$$

$$b = \sum_{k=1}^9 (10 - k) 10_{C_k} = \sum_{k=1}^9 9_{C_k} = 10(2^9 - 1)$$

$$\begin{aligned} 64. \quad A^2 &= B^{-1}AB \\ A^2 &= B^{-1}AB \cdot B^{-1}AB \\ &= (B^{-1})^2 A(B)^2 \\ A^8 &= (B^{-1})^3 A(B)^3 \\ A^7 &= I \end{aligned}$$

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$$

$$A^3 + 3A^2 + 2IA = 0$$

$$\begin{aligned} 65. \quad (A + I)^3 - A &= I^3 \\ A &= (A + I)^3 + (-I)^3 \end{aligned}$$

$$66. \quad AA^T = I \Rightarrow \frac{1}{9} \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{vmatrix} \begin{vmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 2 & -2 & b \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \text{ or}$$

$$\frac{1}{9} \begin{vmatrix} 9 & 0 & a+4+2b \\ 0 & 9 & 2a+2-2b \\ a+4+2b & 2a+2-2b & a^2+4+b^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$67. \quad \text{product of the matrices is } \begin{pmatrix} 1 & an^2 + bn + c \\ 0 & 1 \end{pmatrix}$$

$$t_n = an^2 + bn + c \text{ and}$$

$$68. \quad 378 = an^2 + bn + c \Rightarrow n = 27$$

$$K=25$$

$$69. \quad A^n = \begin{pmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{pmatrix} \text{ and}$$

$$|A^n - I| = 1 - 2^n \Rightarrow \lambda = 2$$

$$70. \quad \Delta = \begin{vmatrix} k & 4 & 1 \\ 2 & 2 & 1 \\ 4 & k & 2 \end{vmatrix} = -(k-4)(k-2) = 0$$

$$\Rightarrow k = 4, 2 \quad \therefore |p| = 12$$

$$24 \left(\frac{1}{|p|} \right) = 24 \times \frac{1}{12} = 2$$

$$71. \quad a = {}^{(2n)}C_{n+1} \text{ and } b = {}^{(2n)}C_{n+1} \text{ then } \frac{a}{b} = 1$$

$$72. \quad AB = I \text{ and trace of } (AB)^r \cdot C_r = 2r \cdot 3^r - 3^r$$

$$2 \sum_{r=1}^{50} r \cdot 3^r - \sum_{r=1}^{50} 3^r = 49 \cdot 3^{51} + 3$$

$$\therefore \frac{a+b}{25} = 4$$

$$73. \quad \sum_{n=1}^6 \operatorname{cosec} \left(\theta + (n-1) \frac{\pi}{4} \right) \operatorname{cosec} \left(\theta + \frac{n\pi}{4} \right)$$

$$= \sqrt{2} \left[\cot \theta - \cot \left(\theta + \frac{3\pi}{2} \right) \right]$$

$$\Rightarrow \cot \theta + \tan \theta = 4 \Rightarrow \theta = 15^\circ \text{ or } 75^\circ$$

$$\therefore \sin^2 \beta - \sin^2 \alpha = \frac{\sqrt{3}}{2}$$

SUBJECTIVE QUESTIONS

74. For non-trivial solution

$$\begin{vmatrix} t-1 & 3t+1 & 2t \\ t-1 & 4t-2 & t+3 \\ 2 & 3t+1 & 3(t-1) \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} t-1 & 3t+1 & 2t \\ 0 & t-3 & 3-t \\ 3-t & 0 & t-3 \end{vmatrix} \begin{matrix} (R_2 \rightarrow R_2 - R_1) \\ (R_3 \rightarrow R_3 - R_1) \end{matrix}$$

$$\Rightarrow (t-3)^2 \begin{vmatrix} t-1 & 3t+1 & 2t \\ 0 & -1 & 1 \\ 1 & 0 & t-3 \end{vmatrix} = 0$$

$$\Rightarrow (t-3)^2 [(t-1)(1) - (3t+1)(-1) + 2t(1)] = 0$$

$$\Rightarrow (t-3)^3 [t-1+3t+1+2t] = 0 \Rightarrow t = 0 \text{ or } t = 3$$

$$\text{giving } a = -2, b = -1. \text{ so, } ab = 2.$$

75. We should have

$$\begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$$\Rightarrow 1(-4k+6) - k(-12+4) + 3(9-2k) = 0$$

$$\Rightarrow -2k + 33 = 0 \Rightarrow k = \frac{33}{2}$$

$$\text{Substituting } k = \frac{33}{2} \text{ and putting } x=m \text{ where}$$

$$m \in Q, \text{ We get the system as}$$

$$\text{There fore, the solution is}$$

$$x = m, y = \frac{-2m}{15}, z = \frac{2m}{5}$$

76. Conceptual

$$77. \quad (A^n - B^n)(A^n + B^n) = A^{2n}B = A(AB) =$$

MATRICES

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$$A(BA) = (AB)A = (BA)A = BA^2$$

similarly $A^3B = BA^3 \dots\dots\dots A^nB = BA^n$.